

ATPL EASA STUDY GUIDE | 090 - COMMUNICATIONS

Chapter 7: Communications Failure & Light Signals

Syllabus Source: CAE Oxford (pp. 169-194) | Pass Target: >= 90% AviationExam

1. ACTION ON SUSPECTED COMMUNICATIONS FAILURE

When two-way radio communication is lost, the flight crew must perform systematic cockpit checks: 1. Check frequency selector, audio control panel switches, volume, squelch, and headset plugs; 2. Attempt contact on previous ATC frequency, company frequency, or 121.500 MHz; 3. If receiver failure is suspected, transmit blind: 'TRANSMITTING BLIND DUE TO RECEIVER FAILURE' (transmit the message twice, including present position, intentions, and time of next transmission); 4. Set transponder Mode A/C code to 7600!

2. RADIO FAILURE PROCEDURES: VMC VS IMC (ICAO ANNEX 2)

The mandatory operational procedure depends strictly on current meteorological flight conditions:

Meteorological Condition	Mandatory ICAO Operational Failure Procedure
Flight in VMC	1. Set squawk 7600; 2. Continue flight in Visual Meteorological Conditions (VMC); 3. Land at the NEAREST SUITABLE AERODROME; 4. Report arrival to ATC by the most expeditious means (telephone).
Flight in IMC (Non-Radar)	1. Set squawk 7600; 2. Maintain assigned speed and level for 20 MINUTES following failure to report over a compulsory reporting point, then conform to filed flight plan; 3. Proceed to destination nav-aid; hold until EAT (or ETA if no EAT); 4. Commence descent at EAT/ETA and land within 30 MINUTES.
Flight in IMC (Radar)	1. Set squawk 7600; 2. Maintain assigned speed and level for 7 MINUTES following 7600 selection or failure to report, then conform to flight plan; 3. Proceed to destination nav-aid, commence descent at EAT/ETA, and land within 30 MINUTES.

[!] EXAM TRAP (AVIATIONEXAM) - AviationExam Critical Numbers: 7 Minutes vs 20 Minutes in IMC

- IN AIRSPACE WHERE RADAR IS USED: Maintain assigned speed and level for 7 MINUTES, then follow flight plan! - IN AIRSPACE WHERE RADAR IS NOT USED: Maintain assigned speed and level for 20 MINUTES, then follow flight plan! - Holding & Descent: Hold at the designated destination navigation aid until the last acknowledged EAT (Expected Approach Time), or ETA if no EAT was given. Land within 30 MINUTES of EAT or ETA!

3. MASTER TABLE OF AERODROME LIGHT SIGNALS FROM CONTROL TOWER

In the event of complete radio failure, the aerodrome control tower directs aircraft using an Aldis lamp (focused optical light gun). Memorize this entire matrix for AviationExam:

Light Signal from Tower	Meaning to Aircraft IN FLIGHT	Meaning to Aircraft ON GROUND
Steady Green	CLEARED TO LAND	CLEARED FOR TAKE-OFF
Flashing Green	RETURN FOR LANDING (Wait for steady green)	CLEARED TO TAXI

Steady Red	GIVE WAY TO OTHER AIRCRAFT & CONTINUE CIRCLING	STOP
Flashing Red	AERODROME UNSAFE, DO NOT LAND	TAXI CLEAR OF RUNWAY IN USE
Flashing White	LAND AT THIS AERODROME & PROCEED TO APRON	RETURN TO STARTING POINT ON AERODROME
Red Pyrotechnic (Flare)	NOTWITHSTANDING PREVIOUS INSTRUCTIONS, DO NOT LAND FOR THE TIME BEING	DO NOT LAND FOR THE TIME BEING

4. AIRCRAFT ACKNOWLEDGEMENT OF TOWER LIGHT SIGNALS

- In Flight (Daylight): Rocking the aircraft's wings (except on final approach).
- In Flight (Night): Flashing landing lights twice, or if not equipped, flashing navigation lights twice.
- On Ground (Daylight): Moving the ailerons or rudder.
- On Ground (Night): Flashing landing lights twice, or flashing navigation lights twice.

5. SSR TRANSPONDER MODES & DATA LINK (MODE S)

Transponder Mode	Data Transmitted to ATC	Operational Role & European Mandate
Mode A	4-digit octal code (4096 combinations)	Provides identity only (e.g. Squawk 7600). Does not transmit altitude.
Mode C	Pressure altitude reporting	Transmits standard pressure altitude referenced to 1013.25 hPa in 100-foot increments.
Mode S	Discrete 24-bit aircraft address	Selective interrogation; altitude reporting in 25-foot increments; downlinks aircraft parameters (DAP: selected altitude, roll angle, IAS/Mach). Mandatory in European airspace.

6. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What does a STEADY RED light signal from the aerodrome control tower mean to an aircraft in flight?

[A] Cleared to land [B] Aerodrome unsafe, do not land [C] Give way to other aircraft and continue circling [D] Return to starting point

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [C]. To an aircraft in flight, Steady Red means 'Give way to other aircraft and continue circling'. (On ground, it means 'Stop').

Q2: What does a FLASHING RED light signal from the control tower mean to an aircraft on the ground?

[A] Stop immediately [B] Taxi clear of runway in use [C] Cleared to taxi [D] Return to starting point

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Flashing Red to an aircraft on the ground means 'Taxi clear of runway in use'. (In flight, it means 'Aerodrome unsafe, do not land').

Q3: What does a FLASHING WHITE light signal from the control tower mean to an aircraft on the ground?

[A] Cleared for take-off [B] Stop [C] Return to starting point on the aerodrome [D] Cleared to taxi

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [C]. Flashing White on the ground instructs the pilot to 'Return to starting point on the aerodrome'.

Q4: An IFR aircraft experiences complete communication failure in IMC while operating in an airspace where radar is not available.

[A] 3 minutes [B] 7 minutes [C] 20 minutes [D] 30 minutes

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. In radar airspace, the pilot maintains assigned speed and level for 7 minutes. In non-radar airspace, the requirement is 20 minutes.

Q5: If an aircraft experiences complete two-way communication failure in VMC, what is the mandatory course of action?

[A] Continue IFR flight plan to original destination [B] Set transponder to 7600, continue in VMC, land at nearest suitable aerodrome, and report to ATC [C] Hold over the nearest VOR for 20 minutes [D] Climb to FL 150

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Under ICAO Annex 2, radio failure in VMC requires maintaining VMC, landing at the nearest suitable aerodrome, and reporting arrival.

Q6: How does an aircraft in flight acknowledge light signals from the control tower during the hours of daylight?

[A] By rocking the aircraft's wings (except on base and final) [B] By firing a green flare [C] By deploying spoilers [D] By cycling the landing gear

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [A]. In daylight, an airborne aircraft acknowledges tower light signals by rocking its wings.

Q7: What does a RED PYROTECHNIC FLARE fired from the control tower signify to an aircraft on final approach?

[A] Cleared to land immediately [B] Notwithstanding any previous instructions, do not land for the time being [C] Windshear warning [D] Runway surface wet

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. A red flare fired from the ground unconditionally forbids landing: 'Notwithstanding any previous instructions, do not land for the time being'.

Q8: What resolution of pressure altitude reporting is provided by Mode S transponders compared to classic Mode C?

[A] Exactly the same (100 ft increments) [B] 25-foot increments (compared to 100 ft in Mode C) [C] 1-foot increments [D] 500-foot increments

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Mode S transponders report pressure altitude in fine 25-foot increments, significantly improving ACAS II and ATC conflict detection compared to Mode C's 100-foot increments.